

SCHOOL OF ENGINEERING

Diploma in Electronics & Computer Engineering (EGDF20)

Wireless Communication & Networking (EGE321)

Tutorial 6: Concepts of Digital Modulation

- 1) The digital string 011010 is to be modulated before transmitting over the air. Draw the modulated signal in the time domain if the following modulation technique is used:
 - (a) Amplitude shift keying
 - (b) Frequency shift keying
 - (c) Phase shift keying
- 2) Determine the theoretical bandwidth efficiency of BPSK, QPSK, 256-QAM and 1024-QAM, 16FSK.
- 3) Calculate the bandwidth efficiency of 8PSK and 8FSK. Which modulation has better bandwidth efficiency?
- 4) QPSK modulation is widely used in digital communication systems. Sketch the vector diagram for
 - i) QPSK modulation and explain the key characteristic.
 - ii) $\pi/4$ QPSK and explain the key characteristic.
- 5) State a figure of merit commonly used to characterise the overall performance of a digital communication system.
- 6) Name two methods of analysing digital modulated signals.
- 7) Name the two types of baseband filters that are widely used in digital communication systems